October 6th, 2026

HAVEN

~ Two Padawans, One Home ~

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Haven

Home should be the easy part

What is it?

Haven is the shared-household app for roommates. Chores, groceries, bills, and rent reminders live in one place, and everyone can see who's doing what.

The Problem

Right now, a household runs on a group chat, a Venmo history, a white board, and everyone's memory. It works until someone forgets, someone's always the one who remembers, and a small thing turns into a big one.

Why?

From previous experience, coordinating household logistics for roommates across several different applications feels disorganized. Having a centralized platform makes it easier to communicate and stay organized with each other.

Intended Users

Who will use the application and what will users be able to accomplish?

- We are targeting roommates sharing an apartment or house together. These are users who deal with shared financial and operational friction every single week.
- Users can track and split bills (Rents, Utilities & Others), manage a shared grocery list that auto-converts to shared expenses, track household chores and receive push notifications as reminders.

What benefit or value will the application provide?

- Roommates currently have to coordinate tasks and logistics through disconnected apps - iMessage, Splitwise, etc
- Instead of managing household related logistics across various apps, we offer a centralized platform for roommates to easily coordinate household chores, groceries, balances to be paid, etc.

What assumptions are you currently making about your users?

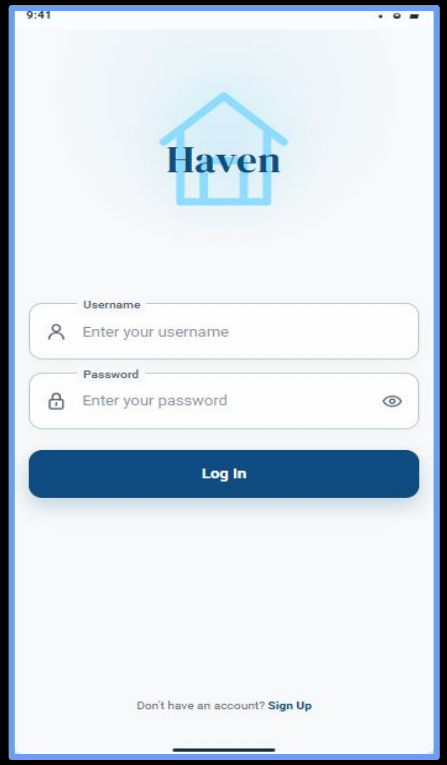
- We assume that the user is living with at least one other person in the household.
- We also assume that the roommates share expenses on a regular basis, and divide household chores among themselves.

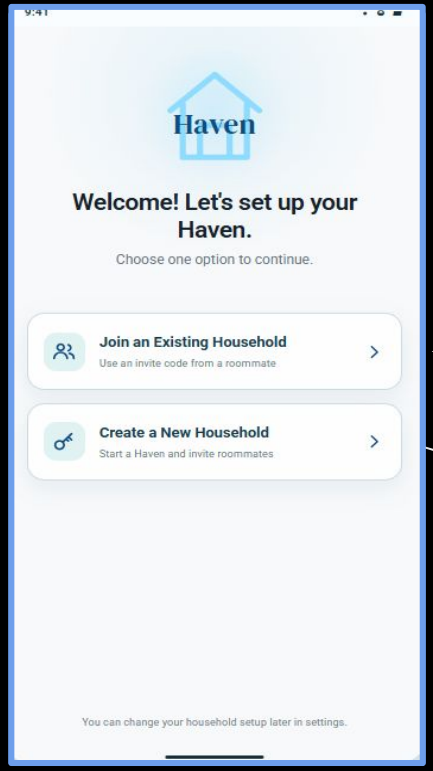
User Experience

*UI is subject to change in future development

Users can sign up and make an account

*If the user already belongs to a household, they will be directed to the household homescreen





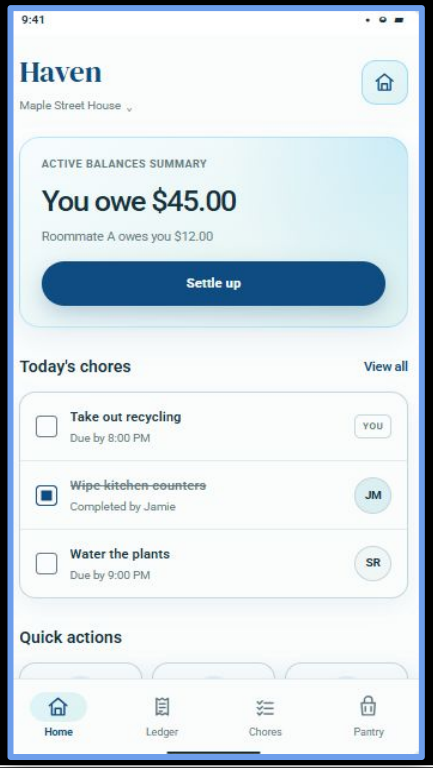
Users can join an existing household

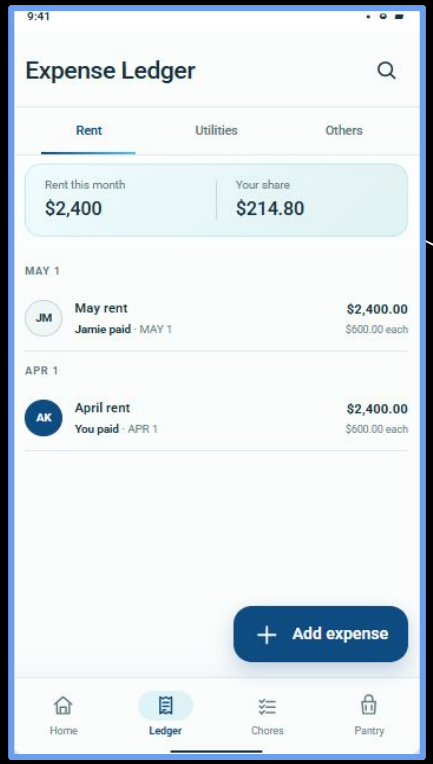
Or create a new household

User Experience

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Dashboard of pending expenses and chores for the user





Segregation of expense type

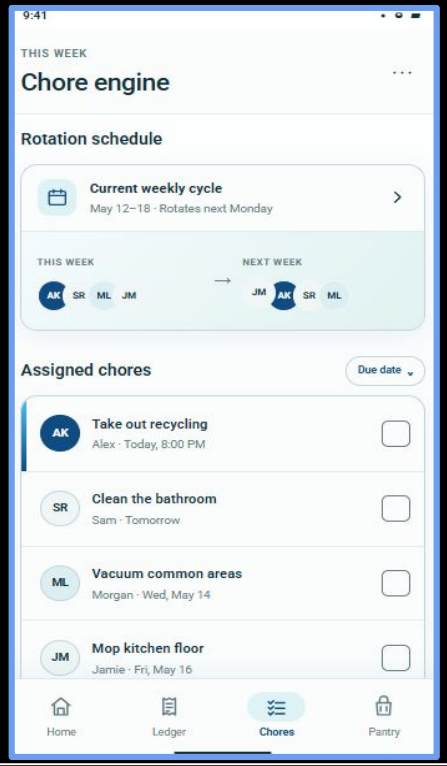
Pending balance reminder

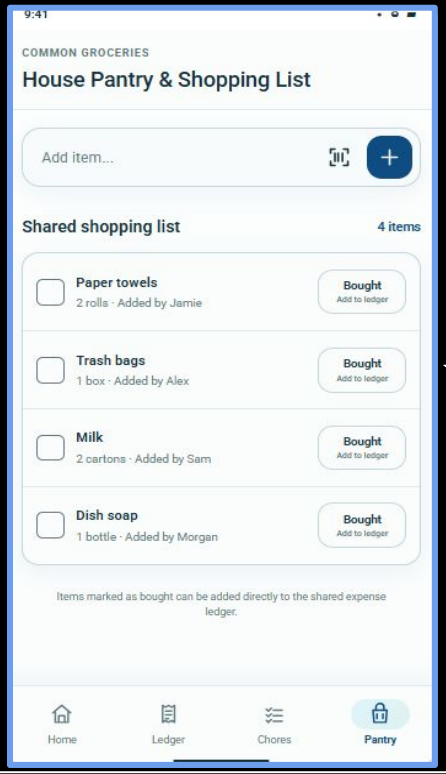
User can add expenses, can also use receipt scanning to digitize receipts

User Experience

*UI is subject to change in future development

Automated assignments for daily and weekly chores





Roommates can add required common household items

User Experience

Minimum Viable Product

- User login
- Creating or joining a household as a roommate
- Log shared expenses
- Chore distribution - weekly and daily
- Common Grocery List
- CRUD operations where applicable for the main features: Exiting a household, deleting user account, etc

Stretch Goals

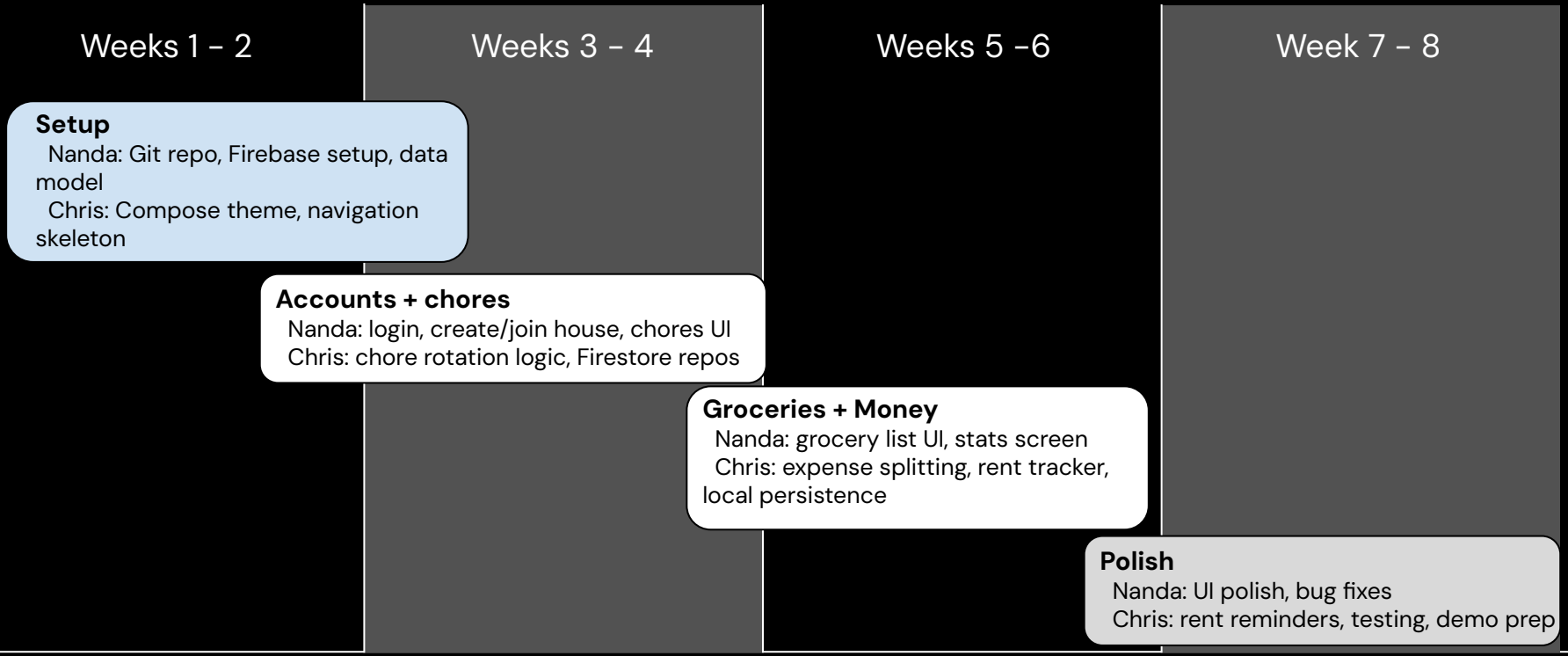
- Push notifications for reminders
- QR code to join a household
- Receipt scanning feature (either paper or screenshot) to digitize paper receipts
- Geospace location sharing, when the user enters a grocery store, a push notification of pending household grocery items can be sent as a reminder to the user

Technical Development

COMPOSE	Kotlin + Jetpack Compose, Material 3
NAVIGATION	Navigation Compose with a 4 tab bottom bar
STATE	MVVM: ViewModel + StateFlow, collected lifecycle-aware
LOCAL DATA STORAGE	Room 
DATABASE + AUTHENTICATION	Firebase Auth, Firestore 
DEVICE	CameraX
PUSH NOTIFICATIONS	Notification Compact library for creating notifications, WorkManager to schedule reminders
THIRD PARTY APIs	Google ML Kit Text Recognition, Google Play services Location API + Google Places API / Places SDK for Android

Developmental Plan

- **Meetings:** twice a week, one to assign tasks and one to check progress
- **Scrum master:** rotates weekly, tracks progress and flags problems early
- **GitHub:** one branch per team member, reviewed and merged into main often
- Proposal and backlog due Oct 20



Technical Risks

- Keeping expenses, balances, and chores, data synchronized correctly across multiple roommates could be challenging. (DataSyncing)
- When a roommate leaves the household or if the user deletes their account, removing their database profile risks breaking old receipts and corrupting past expense history. How do we handle that?
- How should Haven handle user updates made while offline and synchronize them with the server once the device reconnects?
- Calculating accurate expenses and organizing splits (even/uneven), how to handle when new user joins the household?



Questions / Feedback

Thank You

Github Repository https://github.com/nanda002/Haven.git

Generative AI (Claude) was used to brainstorm app name ideas, identify possible APIs and cloud services, and identify any potential risks not accounted for. Generative AI (Figma) was used to create a rough draft of our wireframes

Reference 1 : https://developers.google.com/ml-kit/guides
Reference 2 : https://developer.android.com/develop/background-work/services/alarm
Reference 3 : https://developer.android.com/develop/ui/compose/notifications/create-notification
Reference 4 : https://developers.google.com/maps/documentation/android-sdk/location
Reference 5 : https://developers.google.com/maps/documentation/places/android-sdk/overview